

## FINAL EXPLANATION: PHYSICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

### FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 1.4: ENERGY, WORK, POWER AND MACHINES

Teacher Key: Exemplar Answers (do not give to students)

#### INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the whole unit. Read the driving question: Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails, and where does the energy go when friction makes the nut hot? Answer each section in the space below its prompt. Write in full sentences, show every step of any calculation with units, and use the words and formulae you learned in Lessons 1 to 8. Link every answer back to the mechanic at Gikomba market. Use  $g = 10 \text{ m/s}^2$ . Work on your own.

#### Section 1: The Mechanic at Gikomba and Energy of Motion and Position

**A jua kali mechanic in Gikomba market, Nairobi, fails to loosen a matatu wheel nut with a short spanner. He then succeeds easily with the spanner extended by a metal pipe. An apprentice notices the nut feels warm after the failed attempts.**

**(a) State the two puzzles (mysteries) this scene raised at the start of the unit. (2 marks)**

**(b) Write the formulae for kinetic energy and gravitational potential energy and name the unit of energy. (3 marks)**

**(c) A 70 kg matatu tout runs at 4 m/s. Calculate his kinetic energy. (2 marks)**

**(d) A 5 kg bag of maize is lifted 2 m onto a grain-store shelf in Kitale. Calculate its gravitational potential energy. (2 marks)**

**(e) If the tout doubled his speed to 8 m/s, what would happen to**

**(a)** Puzzle 1: if the spanner-lever system makes the job easier, where does the extra force come from (how does the long spanner give an advantage)? Puzzle 2: if energy is conserved, why does the nut get hot, and why can we never get more work out than we put in? The two puzzles are mechanical advantage and energy dissipation.

**(b)**  $KE = \frac{1}{2}mv^2$  (energy of motion).  $PE = mgh$  (energy stored because of height above a reference point). Energy is measured in joules (J).

**(c)**  $KE = \frac{1}{2} \times 70 \times 4^2 = \frac{1}{2} \times 70 \times 16 = 560 \text{ J}$ .

**(d)**  $PE = mgh = 5 \times 10 \times 2 = 100 \text{ J}$ .

**(e)**  $KE = \frac{1}{2} \times 70 \times 8^2 = \frac{1}{2} \times 70 \times 64 = 2240 \text{ J}$ . Because speed is squared in  $KE = \frac{1}{2}mv^2$ , doubling the speed makes the kinetic energy four times bigger ( $560 \text{ J} \times 4 = 2240 \text{ J}$ ).

his kinetic energy? Explain using the formula. (2 marks)

## Section 2: Work, Force and Displacement

In Lesson 3 you looked at the data from the mechanic's two attempts, using  $W = Fd \cos\theta$  as your model of work. In each row,  $F$  is the force applied and  $d$  is the displacement in the direction of that force ( $\theta = 0^\circ$ ).

| Spanner | Force applied (N) | Displacement (m) |
|---------|-------------------|------------------|
| Short   | 250               | 0                |
| Long    | 80                | 0.45             |

- (a) Write the formula for work, including the angle. (2 marks)
- (b) Use the formula to calculate the work for each row of the table. Show your steps. (4 marks)
- (c) The mechanic pushed hard with the short spanner and got tired. Explain, using the formula, why a scientist says no work was done in the short-spanner row. (2 marks)
- (d) Name the three kinds of work (by sign) that depend on the angle between force and displacement and give the angle for each. Give one example of each from your lessons. (3 marks)

- (a)  $W = Fd \cos\theta$ , where  $F$  is the force,  $d$  is the displacement and  $\theta$  is the angle between the force and the displacement.
- (b) Short spanner row:  $W = 250 \times 0 \times \cos 0^\circ = 0$  J. Long spanner row:  $W = 80 \times 0.45 \times \cos 0^\circ = 80 \times 0.45 \times 1 = 36$  J.
- (c) Work needs displacement. In the short-spanner row the displacement is  $d = 0$ , so  $W = Fd \cos\theta = 0$ , however large the force of 250 N was. The mechanic's muscular energy was still used up, but no useful work was done.
- (d) Positive work:  $\theta = 0^\circ$ , force and motion in the same direction, for example pushing a desk across the floor. Zero work:  $\theta = 90^\circ$ , for example carrying a heavy schoolbag while walking horizontally. Negative work:  $\theta = 180^\circ$ , force opposite to motion, for example friction opposing a sliding desk.

### Section 3: Where Does the Energy Go?

**The apprentice felt that the wheel nut was warm after the mechanic's attempts with the short spanner.**

**(a) Explain why the nut became warm even though it did not move. Name the energy change that took place. (3 marks)**

**(b) In Lesson 4 you pulled a wooden block over smooth tile and over sandpaper. Write the formula for work done against friction and explain what you found about the warmth of the two surfaces and why. (3 marks)**

**(c) Write an equation showing how total work input, useful work output and work done against friction are related. (1 mark)**

**(d) A classmate says: "The heat means energy was destroyed." Use the Law of Conservation of Energy to explain why this is wrong. (3 marks)**

(a) Friction acts where the spanner touches the nut and where the nut touches the bolt thread. The mechanic's muscular (chemical) energy was converted by friction into thermal energy, so the nut became warm. Energy change: muscular energy → work against friction → thermal energy.

(b)  $W_f = f_r \times d$ , where  $f_r$  is the friction force in newtons and  $d$  is the distance slid in metres. The sandpaper felt warmer than the smooth tile after repeated pulls. Rougher surfaces give a larger friction force, so for the same distance more work is done against friction and more thermal energy is produced.

(c)  $W_{in} = W_{out} + W_{friction}$ .

(d) Energy cannot be created or destroyed, only changed from one form to another. The energy that the mechanic put in and that did not become useful work was transformed into thermal energy in the nut and spanner and then spread into the metal. It is not lost. It is simply no longer useful for doing work.

### Section 4: How the Long Spanner Helps: Machines, MA, VR and Efficiency

**The spanner and its pipe extension work as a lever.**

**(a) Write the formulae for mechanical advantage (MA), velocity ratio (VR) and efficiency. (3 marks)**

**(b) Explain how the long spanner changes the effort needed to loosen the nut. Say what happens to the distance the mechanic's hand moves compared with the distance the load moves. (3 marks)**

**(c) A classmate says: "The long**

(a)  $MA = \text{Load} \div \text{Effort}$ .  $VR = \text{effort distance} \div \text{load distance}$ .  $\text{Efficiency (\%)} = (MA \div VR) \times 100\%$ .

(b) The pipe makes the effort arm longer, which increases the mechanical advantage, so a smaller effort (force) can overcome the nut's resistance. The cost is that the mechanic's hand must move through a larger distance than the load (the nut's edge) moves. The machine trades force for distance.

(c) I disagree. No machine creates extra energy. The work the mechanic puts in is shared between useful work on the nut and work done against friction ( $W_{in} = W_{out} + W_{friction}$ ), so the work out can never be greater than the work in. The long spanner only redistributes force and distance: a smaller effort over a larger effort distance does the job that a larger force over a smaller distance would otherwise do.

(d) In an ideal machine with no friction, work out equals work in and MA equals VR. In a real machine, friction in the nut threads and at the spanner joints changes some of the work input into heat. The useful work output is therefore less than the total work input, so efficiency ( $MA \div VR \times 100\%$ ) is below 100%.

|                                                                                                                                                                                                                                             |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>spanner gives the mechanic extra energy." Use the idea of work in and work out to say whether you agree. (3 marks)</p> <p>(d) Explain why the efficiency of any real machine, including the spanner, is always below 100%. (2 marks)</p> |  |
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| Section 5: The Full Story and Power                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p><b>Now put the whole unit together to answer the driving question: Why can the jua kali mechanic loosen the wheel nut effortlessly with a long spanner when a short one fails, and where does the energy go when friction makes the nut hot?</b></p> <p><b>Write a clear explanation of about 6 to 8 sentences. Use at least four of these terms correctly: work, mechanical advantage, velocity ratio, friction, thermal energy, efficiency, conservation of energy, power.</b></p> <p><b>Then answer: Two mechanics loosen the same nut and do the same work, but one takes less time. Which one has the greater power? Write the formula for power and its unit. (Total: 8 marks for the explanation, 3 marks for power)</b></p> | <p>The long spanner extension makes the effort arm longer, so the lever has a greater mechanical advantage. This means the mechanic needs a smaller force, but his hand moves through a larger distance than the load, so the machine trades force for distance. It does not create extra energy, because work in equals useful work out plus work done against friction (<math>W_{in} = W_{out} + W_{friction}</math>). With the short spanner there was no displacement in the data (<math>d = 0</math>), so no work was done in that row of the table, yet his muscular energy was still used up. Friction at the contact surfaces changed that energy into thermal energy, which is why the nut felt warm. The energy was not destroyed, because energy is always conserved, but it became thermal energy that spreads into the metal and cannot do useful work. Because of this friction, the work out is always less than the work in, so the efficiency of any real machine is always less than 100%.</p> <p>Power: the mechanic who takes less time to do the same work has the greater power. <math>P = W/t</math>, measured in watts (W), where <math>1 W = 1 J/s</math>. Power measures how fast work is done, not how much work is done.</p> |

| RUBRIC                                          |                                                                                                                                                                   |                                                                                                                                                               |                                                                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Criterion                                       | Excellent (4)                                                                                                                                                     | Proficient (3)                                                                                                                                                | Developing (1–2)                                                                                                    |
| Calculations and use of formulae (KE, PE, work) | All calculations are correct, with formulae stated, steps shown and units given. The meaning of $d = 0$ and the effect of doubling speed are explained correctly. | Most calculations are correct with formulae and units. There may be a small slip in a step or a unit, or the explanation of zero work or of doubling speed is | Formulae are missing or wrongly used. Calculations have several errors or no units, and zero work is not explained. |

|                                                                      |                                                                                                                                                                                                                 |                                                                                                                                                   |                                                                                                              |
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|                                                                      |                                                                                                                                                                                                                 | incomplete.                                                                                                                                       |                                                                                                              |
| Explaining energy transformation and dissipation (friction and heat) | Clearly explains that friction changes muscular energy into thermal energy, uses $W_{in} = W_{out} + W_{friction}$ , and correctly argues that energy is conserved and not destroyed.                           | States that friction produces heat and that energy is conserved, but the link to work against friction or the equation is thin or partly missing. | Says energy is lost or destroyed, or gives no mechanism for the warm nut.                                    |
| Explaining machines, MA, VR and efficiency                           | Explains that the longer effort arm raises MA and that force is traded for distance. States that no machine creates energy and that efficiency is below 100% because friction makes work out less than work in. | Explains the lever advantage correctly but is vague about the force-distance trade or about why efficiency is below 100%.                         | Suggests the machine gives extra energy, or cannot state the MA, VR or efficiency formulae.                  |
| Integrated explanation of the phenomenon and use of vocabulary       | A coherent, linked explanation of both puzzles that uses at least four unit terms correctly, with power distinguished as rate of working ( $P = W/t$ , watts).                                                  | A mostly connected explanation using four terms, with minor misuse, or a power answer that is correct but incomplete.                             | A list of facts that does not link to the mechanic, uses fewer than four terms, or confuses power with work. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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